EXPERIMENT 02

Aim: Prepare a detailed Software Requirement Specification (SRS) document for the Employee Management System as per IEEE standard.

Project Title: Employee Management System

1. Introduction

1.1 Purpose

This document describes the Software Requirements Specification (SRS) for the **Employee Management System (EMS)**. The system is designed to automate employee-related operations such as employee record management, attendance tracking, leave management, payroll processing, department management, and performance evaluation. It enables employees to access their personal information and HR services while allowing administrators and HR managers to efficiently manage employee data and organizational processes.

1.2 Scope

The Employee Management System provides a centralized platform for managing employee information within an organization. The system allows administrators to maintain employee records, monitor attendance, manage payroll, approve leave requests, and generate reports. Employees can update their profiles, apply for leave, view attendance records, and download salary slips. The system reduces manual work, improves data accuracy, enhances security, and increases overall organizational efficiency.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS:** Software Requirements Specification
- **EMS:** Employee Management System
- **HR:** Human Resources
- **Admin:** System Administrator responsible for managing employees and system operations
- **Employee:** Registered user of the system
- **DB:** Database
- **UI:** User Interface
- **API:** Application Programming Interface

1.4 References

- IEEE 830-1998 Software Requirements Specification Standard
- Organization Human Resource Policies
- Software Engineering by Ian Sommerville
- Employee Management System Project Documentation

1.5 Overview

This SRS describes the overall description, functional requirements, non-functional requirements, constraints, assumptions, interfaces, database requirements, and supporting information for the Employee Management System. It serves as a reference for developers, testers, project managers, and stakeholders during the software development life cycle.

2. Overall Description

2.1 Product Perspective

The Employee Management System is a standalone web-based application connected to a centralized database. It provides secure access to administrators, HR managers, and employees through role-based authentication. The system automates HR operations and minimizes manual record keeping.

2.2 Product Functions

The system provides the following major functions:

- Employee Registration
- Employee Profile Management
- Attendance Management
- Leave Management
- Payroll Management
- Department Management
- Performance Evaluation
- User Authentication
- Report Generation
- Dashboard Analytics

2.3 User Classes and Characteristics

Administrator

- Manage employee records
- Manage departments
- Process payroll
- Generate reports
- Manage user accounts

HR Manager

- Add and update employee information
- Approve or reject leave requests
- View attendance reports
- Conduct performance evaluations

Employee

- Login securely
- Update personal profile
- Apply for leave
- View attendance history
- Download salary slips

2.4 Design and Implementation Constraints

- Secure login with authentication
- Role-based access control
- Responsive web interface
- Database integrity
- Secure communication using HTTPS

2.5 Assumptions and Dependencies

- Internet connection is available.
- Users have valid login credentials.
- Database server remains operational.
- Email services are configured for notifications.

3. Specific Requirements

3.1 Detailed Functional Requirements (Working Criteria)

- Admin can securely log in.
- HR can add, edit, and delete employee records.
- Employees can log in and manage their profiles.
- Employees can mark attendance.
- Employees can apply for leave.
- HR can approve or reject leave requests.
- The system generates payroll and salary slips.
- Admin can manage departments.
- The system generates employee, attendance, leave, and payroll reports.
- All employee data is securely stored in the database.

3.2 Non-functional Requirements

- **Performance:** Response time within 3 seconds.
- **Storage:** Securely stores employee and payroll data.
- **Security:** Password encryption and role-based access.
- **Usability:** Easy-to-use interface.
- **Reliability:** High system availability.
- **Interoperability:** Supports major web browsers.
- **Scalability:** Supports future expansion.
- **Backup:** Regular database backup and recovery.

3.3 Interfaces

- **User Interface:** Login, Dashboard, Employee, Attendance, Leave, Payroll, and Reports pages.
- **Hardware Interface:** Standard computer or laptop with internet.
- **Software Interface:** Windows/Linux, MySQL, Apache/Nginx.
- **Communication Interface:** HTTP/HTTPS, SMTP email, REST APIs.

4. Appendices

- Sample Employee Registration Form
- Sample Leave Application Form
- Sample Salary Slip
- ER Diagram
- Use Case Diagram
- System Flowchart
- Glossary of Terms